

Exercise 4 – CPU Scheduling

Questions are taken from Stallings, Operating Systems Internals and Design Principles, fifth edition and Silberschatz et al., Operating System Concepts, seventh edition.

1 – Silberschatz 5.4

Consider the following set of processes, with the length of the CPU burst given in milliseconds.

Process	Burst Time	Priority
P_1	10	3
P_2	1	1
P_3	2	3
P_4	1	4
P_5	5	2

The processes are assumed to have arrived in the order P_1, P_2, P_3, P_4, P_5 all at time 0.

- Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1).
- What is the turnaround time of each process for each of the scheduling algorithms in part a?
- What is the waiting time of each process for each of the scheduling algorithms in part a?
- Which of the algorithms om part a results in the minimum average waiting time (over all processes)?

2

Consider the following set of processes, with the length of the CPU burst and arrival time given in milliseconds.

Process	Burst Time	Arrival time
P_1	8	0
P_2	4	0.4
P_3	1	1

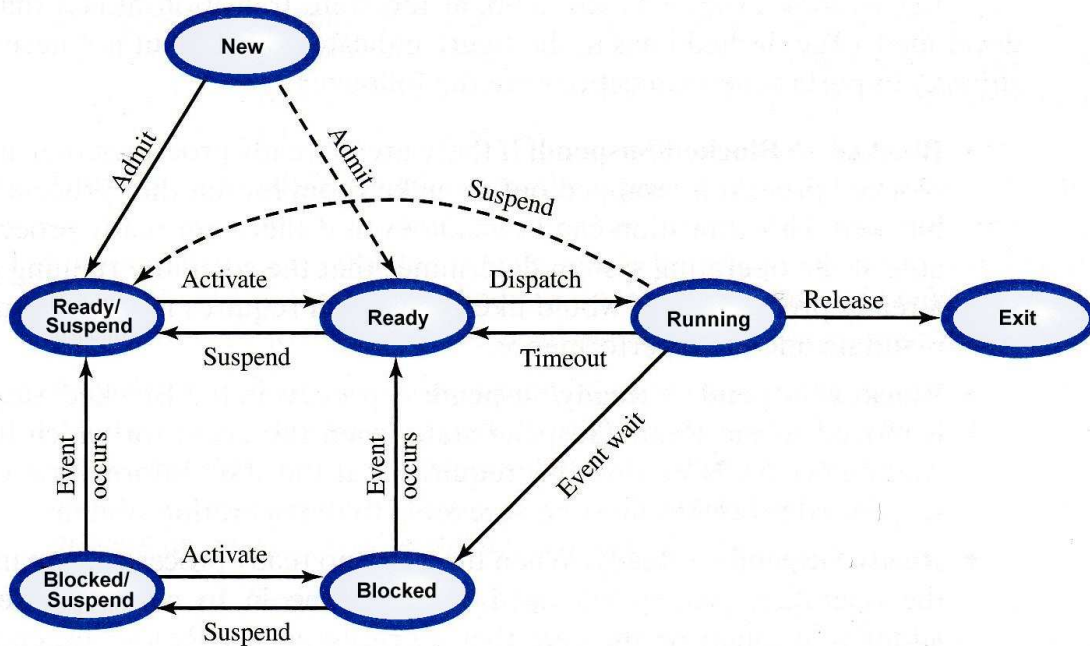
- Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, Clairvoyant SJF (the algorithm can look into the future and wait for a shorter process that will arrive).
- What is the turnaround time of each process for each of the scheduling algorithms in part a?
- What is the waiting time of each process for each of the scheduling algorithms in part a?
- Which of the algorithms om part a results in the minimum average waiting time (over all processes)?

3 – Stallings 3.4

Consider the state transition diagram in the figure. Suppose that it is time for the operating system to dispatch a process and that there are processes in both the Ready state and the Ready/Suspend state, and that at least one process in the Ready/Suspend state has higher scheduling priority than any of the processes in the Ready state. Two extreme policies are:

1. Always dispatch from a process in the Ready state, to minimize swapping.
2. Always give preference to the highest-priority process, even though that may mean swapping when swapping is not necessary.

Suggest an intermediate policy that tries to balance the concerns of priority and performance.



- Ready: The process is in main memory and available for execution.
- Blocked: The process is in main memory and awaiting an event.
- Blocked/Suspend: The process is in secondary memory and awaiting an event.
- Ready/Suspend: The process is in secondary memory but is available for execution as soon as it is loaded into main memory.

4 – Stallings 3.9

In a number of early computers, an interrupt caused the register values to be stored in fixed locations associated with the given interrupt signal. Under what circumstances is this a practical technique? Explain why it is inconvenient in general.